
Sorting Things into Groups (Classification)

Contact Session 4 — Module 4 (Part 1)

Drawing the line between “yes” and “no.” What a decision boundary is, what straight and curvy boundaries look like, and the simple rule for deciding which side a new point is on.

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|------------------|------------------|
| ■ Intuition | ■ Analogy |
| ■ Worked example | ■ Solved problem |
| ■ Common pitfall | ■ Key takeaways |

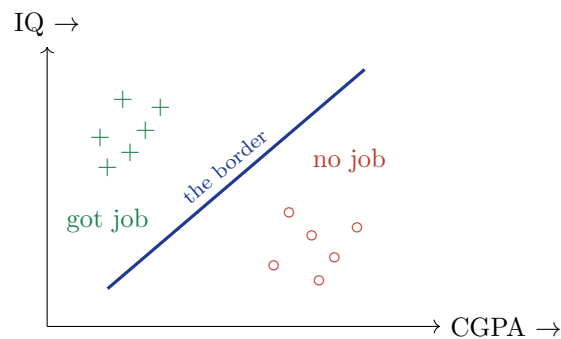
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1 From “how much?” to “which one?”

Last session we guessed a *number* (a price). Now we guess a *group*: spam or not spam, will-get-the-job or won’t, which digit 0–9. The answer is no longer a number on a scale — it’s a label from a short list. This is **classification**.

The neatest way to picture it: put each example as a dot on a graph, using its facts as the position. Dots of one group cluster in one area, dots of the other group cluster elsewhere. Our job is to **draw the border between the clusters**.



Once we have the border, classifying a new person is easy: see which side of the border they land on. The border is called the **decision boundary**, and the two areas it cuts the graph into are the **decision regions**.

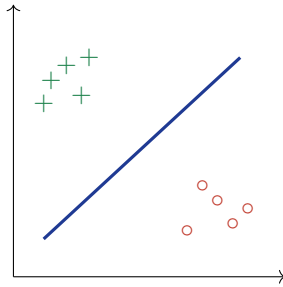
★ Regression draws *through*, classification draws *between*

In regression we drew a line *through* the dots to predict a height. In classification we draw a line *between* the dots to split them into camps. Same idea of “draw a helpful line,” but now the line is a fence, not a trend.

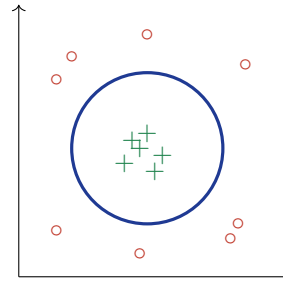
2 Straight fences and curvy fences

This is the heart of the session, so let's go slowly. The border between groups can be **straight** or **curvy**, and which one you need depends on how the dots are arranged.

A straight fence is enough



Here you need a curve



On the **left**, the two groups sit in tidy clumps, so a single *straight* fence separates them perfectly. We call this a **linear boundary**, and a method that can only draw straight fences is a **linear classifier**.

On the **right**, one group is trapped *inside* the other — a ring of circles around a core of pluses. No straight line can split those. You need a *curvy* fence, a **non-linear boundary**.

~> Slicing with a knife vs. a cookie cutter

A straight fence is a single straight knife-cut across a board — great when the two kinds of crumbs are on opposite ends. But if one kind is piled in the *middle* surrounded by the other, one straight cut can't separate them; you need a round cookie-cutter (a curve) to scoop the middle out. The shape of the fence has to match the shape of the mess.

★ The good news: curves come almost for free

Remember the trick from Module 3? Feed the model *extra made-up facts* like “CGPA squared” or “CGPA $\times$ IQ,” and a straight-fence method can suddenly bend its fence into a curve — because a straight fence in the bigger fact-space looks curvy back in the original picture. So even a “linear” classifier can draw curves once you feed it the right ingredients. We meet the plain straight-fence version first because it's the easiest to understand.

3 The simple rule for picking a side

How does a straight-fence classifier actually decide? It does the same “multiply-and-add” we already know. It takes the facts, multiplies each by a weight, adds them up, and gets a single *score*:

$$\text{score} = w_0 + w_1 \cdot \text{fact}_1 + w_2 \cdot \text{fact}_2 + \dots$$

Then it just checks the sign:

- score **above zero** $\Rightarrow$ one group,
- score **below zero** $\Rightarrow$ the other group.

The border itself is exactly where the score equals *zero* — and “score = 0” for a multiply-and-add rule is always a straight line (or a flat sheet, in higher dimensions).

★ Why “add up the facts” makes a *straight* fence

Think of the score as a tilted ramp laid over the graph: it rises as you move one way and falls the other. The places where the ramp is exactly at “sea level” (score = 0) form a perfectly straight line. Everything uphill is one group, everything downhill is the other. That’s why plain multiply-and-add can only ever draw straight fences — the ramp is flat.

Σ Which side is this person on?

A classifier scores job offers with $\text{score} = 0.4 + 0.3 \text{ CGPA} - 0.45 \text{ IQ}$. For a candidate with CGPA=5, IQ=6:

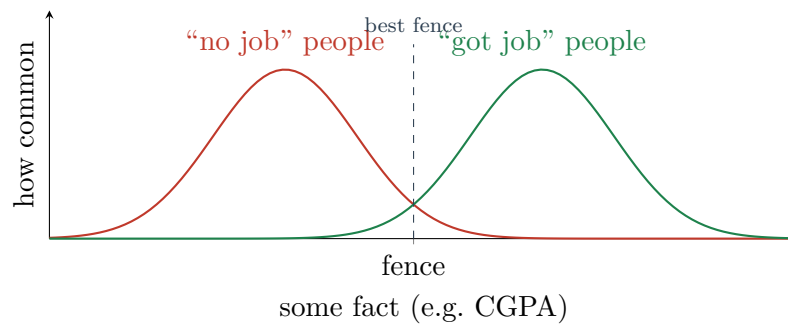
$$\text{score} = 0.4 + 0.3(5) - 0.45(6) = 0.4 + 1.5 - 2.7 = -0.8.$$

The score is below zero, so this person lands on the “no job” side. The fence itself is the line where $0.4 + 0.3 \text{ CGPA} - 0.45 \text{ IQ} = 0$.

4 When groups overlap: choosing the smart way

In real life the two clumps usually *overlap* a bit — some high-CGPA people still don't get the job. So no fence is perfect. What's the *best* we can do?

The sensible rule: for any new point, ask “which group is *more likely* right here?” and go with that one. This is the **best-guess rule** (textbooks call it the Bayes rule).



The two humps are the two groups. Where they overlap (the middle), some mistakes are simply *unavoidable* — both groups have people there. The best place for the fence is exactly where the two humps cross, because that's where “more likely” switches from one group to the other.

! Common Pitfall

Sliding the fence away from the crossing point always makes things *worse* overall: you rescue a few points on one side but lose more on the other. (The one exception is when one mistake is far costlier than the other — e.g. missing a disease vs. a false alarm — and then you deliberately shift the fence. More on that next session.)

5 Two styles of classifier (good to know the names)

There are two broad ways to build a classifier, and the difference is simple.

Learn each group fully
then ask “which group
more likely made this?”
(*generative*)

Learn only the border
straight to “which side is it on?”
(*discriminative*)

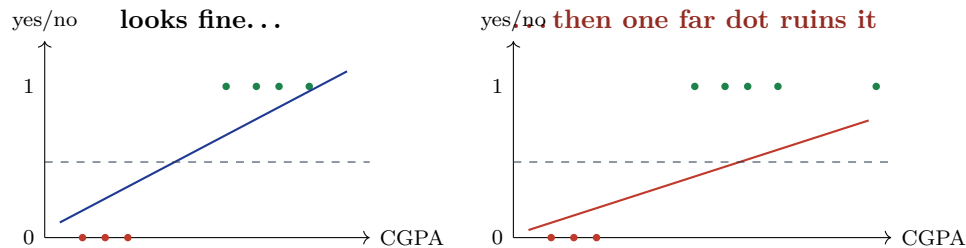
↪ Telling English from French

Learn each group fully: study English *and* French completely, then for a new sentence ask “which language was more likely to produce this?” **Learn only the border:** don’t study either language — just learn the giveaway differences (“*the*” → English, “*le*” → French). The border-only approach is lazier but often wins, because it spends all its effort exactly where the decision is made.

Logistic regression — our star next session — is the “learn only the border” kind.

6 Why we don't just reuse the line from Module 3

Tempting idea: label “yes” as 1 and “no” as 0, fit a straight line like in regression, and call anything above 0.5 a “yes.” It sort of works... until it doesn't.



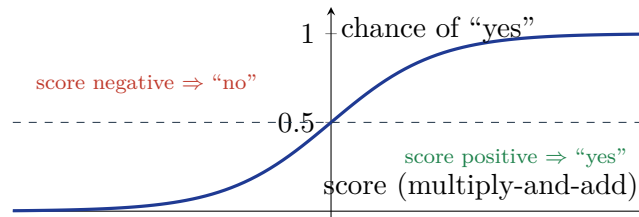
Two problems. First, a straight line can shoot *above 1* or *below 0*, but a yes/no answer should behave like a *chance* between 0 and 1 — “above 1 likely” is meaningless. Second, one far-away “obvious yes” dot drags the whole line down and shoves the 0.5 mark sideways, suddenly mislabelling points that were fine before.

★ Intuition

A straight line tries to *hit* the targets 0 and 1 exactly, so a dot that's already “super clearly yes” (far to the right) still creates a big pull, bending the fence out of place. We want something that once a point is clearly on one side, *stops caring* about how far. That something is coming next.

7 The fix in one picture: the S-curve

Instead of a straight line, we pass the score through a gentle **S-shaped curve** (its real name is the *sigmoid*). The S-curve takes any score — huge negative to huge positive — and squashes it into a chance between 0 and 1.



Read the curve: a very negative score gives a chance near 0 (“almost certainly no”), a very positive score gives a chance near 1 (“almost certainly yes”), and a score of exactly 0 gives 0.5 (“a coin flip”). The fence is still where the score is 0 — so it’s still a *straight* fence — but now the output is a sensible chance, and far-away dots no longer yank it around (the curve has already flattened out there).

The S-curve as a formula. The curve has a tidy formula. If the score is z , the chance is

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Don’t memorise it — *read* it, and watch the whole S appear. Here e^{-z} is just a positive number that shrinks as z grows. So:

- big positive score: $e^{-z} \rightarrow 0$, so $\sigma \rightarrow \frac{1}{1+0} = 1$ (almost certainly yes).
- big negative score: $e^{-z} \rightarrow \text{huge}$, so $\sigma \rightarrow \frac{1}{\text{huge}} = 0$ (almost certainly no).
- score of 0: $e^0 = 1$, so $\sigma = \frac{1}{2}$ (the coin flip, right on the fence).

That one little fraction does exactly the job we wanted: any score goes in, a sensible chance between 0 and 1 comes out.

↪ A dimmer switch, not an on/off switch

A hard rule is a light switch: off below the fence, fully on above. The S-curve is a *dimmer*: it slides smoothly from off to on, so near the fence it can say “55% yes,” and far from the fence it settles at “basically 0” or “basically 1” and stops reacting. That calm at the edges is exactly why it ignores the far-away dot that wrecked the straight line.

Squashing the multiply-and-add score through this S-curve gives us **logistic regression** — how we actually train it (and how it reads off real probabilities) is the whole of the next session.

8 The whole story on one page

★ Key Takeaways

- Classification puts each example in a **group**; on a graph, that means **drawing a border** (decision boundary) between clusters.
- A **straight border** (linear) works when the groups sit in separate clumps; a **curvy border** (non-linear) is needed when one group wraps around another.
- Feeding in extra made-up facts (squares, products) lets a straight-fence method bend into curves — so “linear” isn’t as limited as it sounds.
- A straight-fence classifier **adds up the facts into a score** and checks the sign; the border is where the score is zero.
- When groups overlap, use the **best-guess rule**: pick whichever group is more likely there; the best fence sits where the groups cross.
- Plain regression makes a poor classifier (it can output silly “chances” and one far dot ruins it); the **S-curve** fixes both, giving **logistic regression** (next session).

Where this comes from: Bishop, *Pattern Recognition & Machine Learning*, Ch. 3–4. Course slides M4 (BITS Pilani WILP).